```
import pandas as pd
```

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Revision history

Notebook info

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```
ain_test_split
lEncoder, StandardScaler

rRegression
egressor

ed_error, r2_score
```

```
_dataset (1).csv")
```

```
_Used(tons)','Pesticide_Used(kg)',

rrors='coerce')
```

```
'Crop_Type'])
rm(df['Irrigation_Type'])
'Soil_Type'])
df['Season'] = le.fit_transform(df['Season'])
```

```
X = df.drop(['Farm_ID', 'Yield(tons)'], axis=1)
y = df['Yield(tons)']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

mse_lr = mean_squared_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)

print("Linear Regression -> MSE:", mse_lr, " R2:", r2_lr)
```

```
Linear Regression -> MSE: 159.58778678263377  R2: 0.0005909065033985472
```

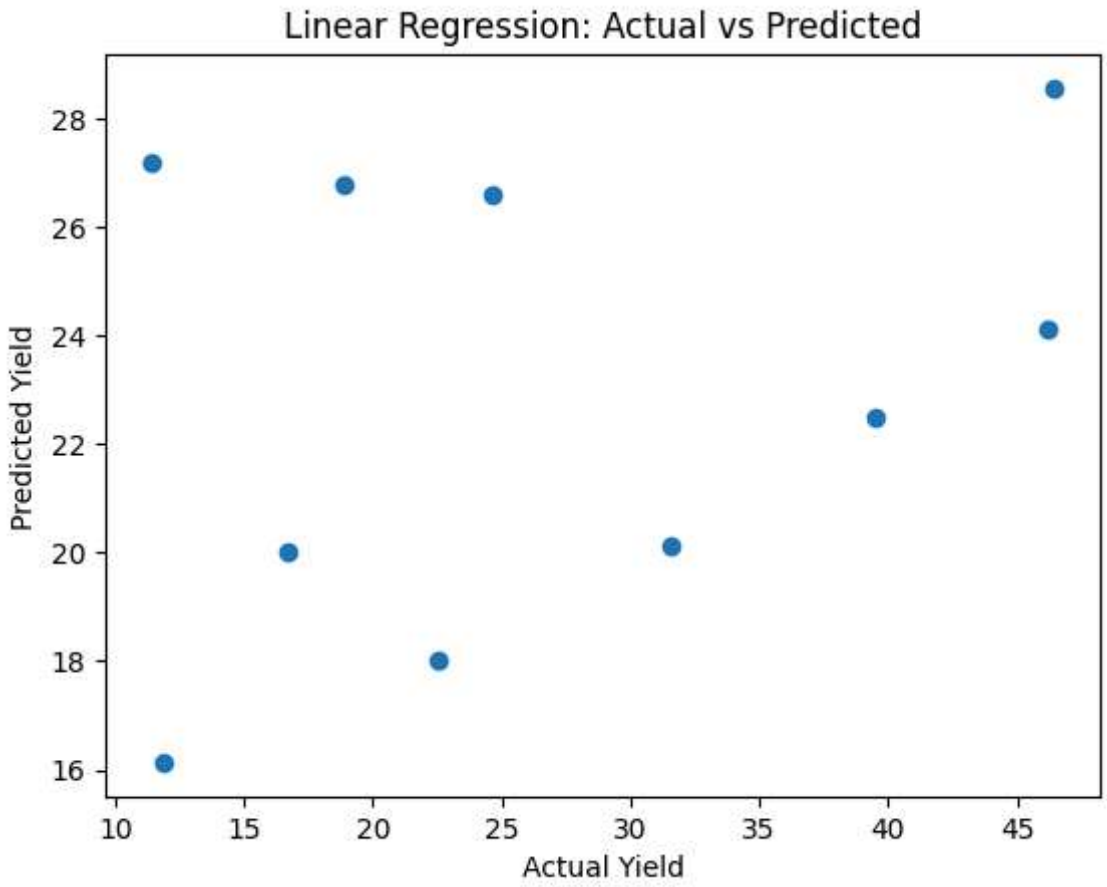
```
dt = DecisionTreeRegressor(random_state=42)
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)

mse_dt = mean_squared_error(y_test, y_pred_dt)
r2_dt = r2_score(y_test, y_pred_dt)

print("Decision Tree -> MSE:", mse_dt, " R2:", r2_dt)
```

```
Decision Tree -> MSE: 390.43117  R2: -1.4450521531073632
```

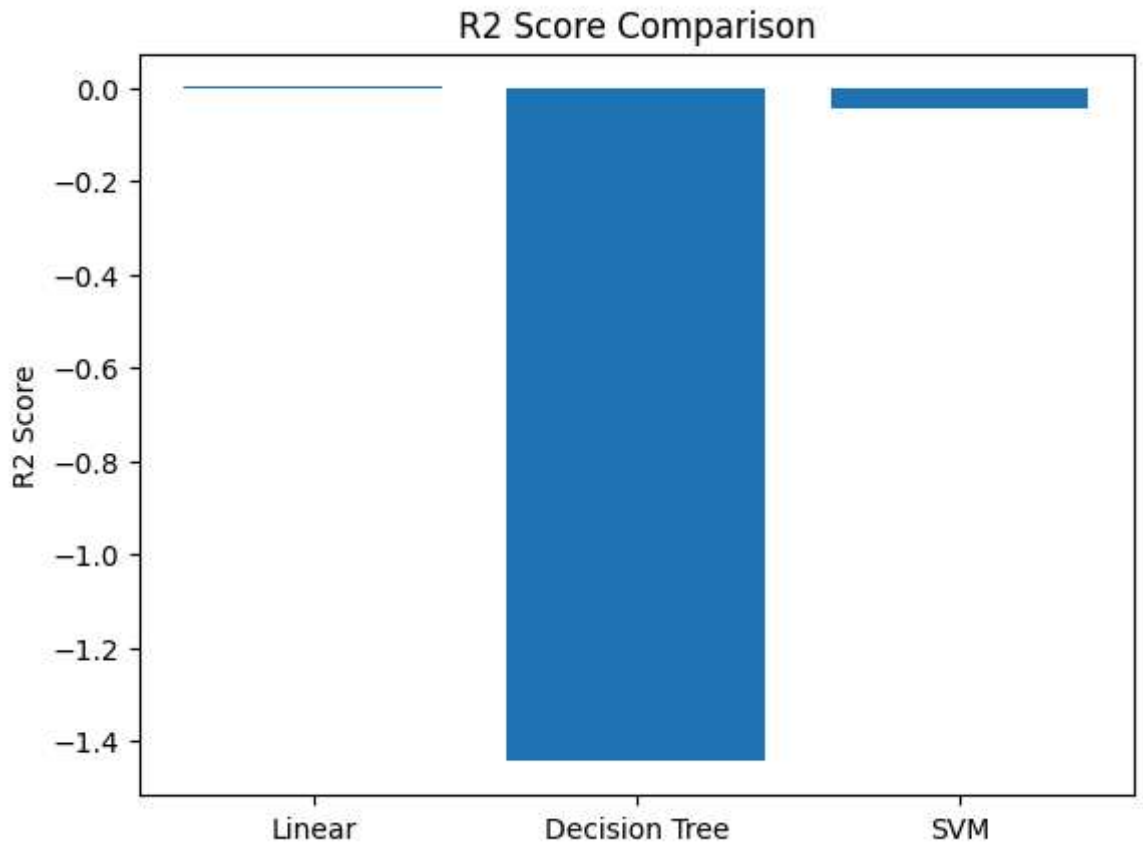
```
svm = SVR()
svm.fit(X_train_scaled, y_train)

y_pred_svm = svm.predict(X_test_scaled)

mse_svm = mean_squared_error(y_test, y_pred_svm)
r2_svm = r2_score(y_test, y_pred_svm)

print("SVM -> MSE:", mse_svm, " R2:", r2_svm)
```

```
SVM -> MSE: 166.98280884147528  R2: -0.045719982576607254
```

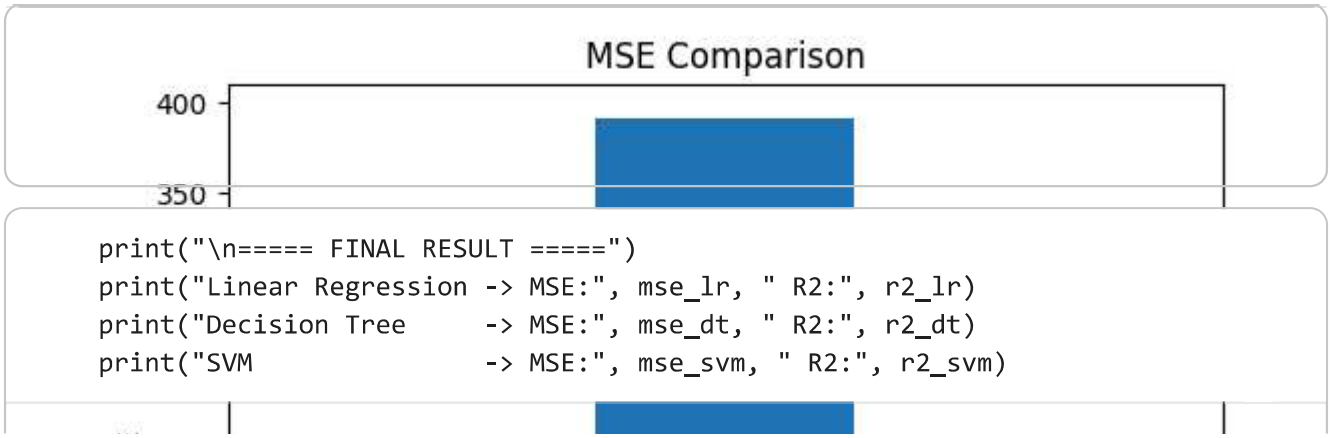
```
plt.figure()
plt.scatter(y_test, y_pred_lr)
plt.xlabel("Actual Yield")
plt.ylabel("Predicted Yield")
plt.title("Linear Regression: Actual vs Predicted")
plt.show()
```




```
models = ['Linear', 'Decision Tree', 'SVM']
r2_scores = [r2_lr, r2_dt, r2_svm]

plt.figure()
plt.bar(models, r2_scores)
plt.title("R2 Score Comparison")
plt.ylabel("R2 Score")
plt.show()
```




```
mse_scores = [mse_lr, mse_dt, mse_svm]

plt.figure()
plt.bar(models, mse_scores)
plt.title("MSE Comparison")
plt.ylabel("MSE")
plt.show()
```




```
print("\n===== FINAL RESULT =====")
print("Linear Regression -> MSE:", mse_lr, " R2:", r2_lr)
print("Decision Tree     -> MSE:", mse_dt, " R2:", r2_dt)
print("SVM               -> MSE:", mse_svm, " R2:", r2_svm)
```